

EXPERIMENT 5

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Experiment -5

Implementation of Dynamic Routing using RIP (Routing Information Protocol)

Aim

To implement Dynamic Routing using RIP (Routing Information Protocol) in Cisco Packet Tracer and understand how routers automatically learn and update routing information between different networks.

Objective

- Deploy Routing Information Protocol (RIP) within Cisco Packet Tracer to observe dynamic path determination.
- Examine the mechanics of distance-vector routing and automatic routing table updates.
- Execute Cisco IOS command-line sequences (router rip, version 2, network, no auto-summary) across connected gateways.
- Evaluate hop count as the primary path metric, noting its 15-hop operational limit.
- Validate cross-subnet data transmission and end-to-end ICMP reachability using ping and traceroute.
- Inspect dynamically learned entries marked with R via the show ip route command.

Software Requirements

- Cisco Packet Tracer

Hardware (Virtual)

- Routers
- Switches

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- PCs
- Copper Straight-Through Cables

Theory

Routing functions as the fundamental process of selecting optimal paths to forward data packets across disparate network segments. Because devices residing on separate subnets cannot directly communicate at Layer 2, routers step in to bridge these boundaries. Dynamic routing modernizes this process by enabling networking hardware to autonomously exchange topological data and adjust path selections without requiring manual administrator updates.

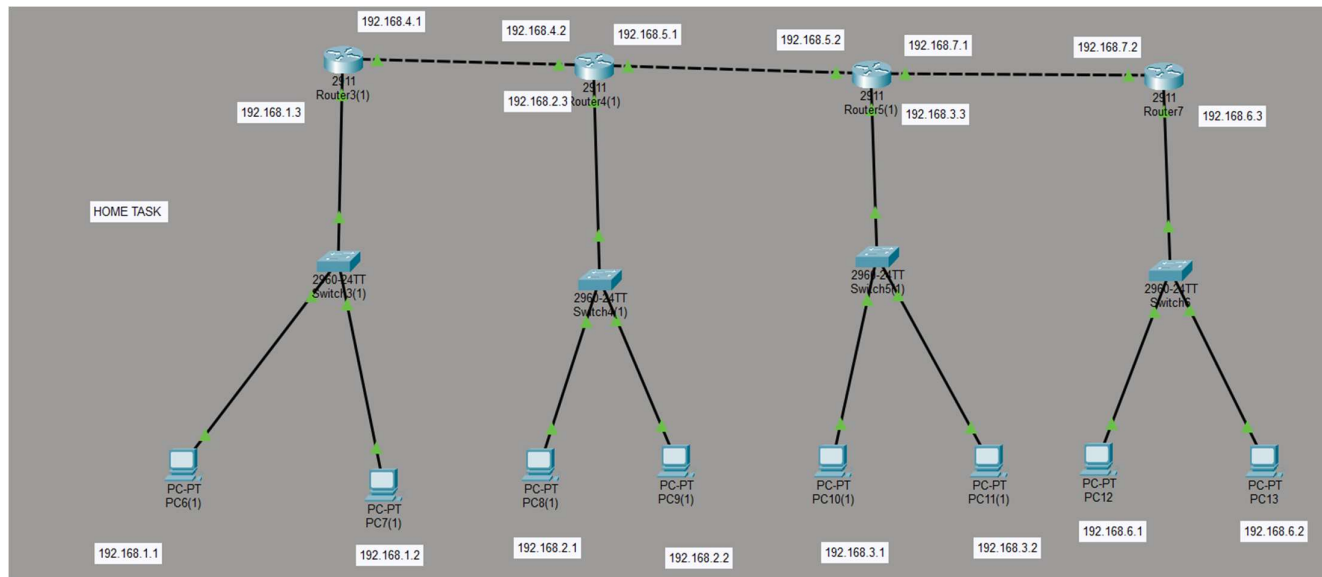
Routing Information Protocol (RIP) operates as a straightforward distance-vector protocol designed to facilitate this dynamic exchange between neighboring devices. Path selection in RIP relies entirely on hop count, defined as the number of intermediate routers a packet must traverse to reach its destination. To maintain stability and prevent infinite routing loops, the protocol enforces a strict upper limit of 15 hops, treating any route that reaches 16 hops as completely unreachable.

Practical / Experiment Steps

1. Open Cisco Packet Tracer.
2. Place the required routers, switches, and PCs in the workspace.
3. Connect the PCs to the switches.
4. Connect the switches to the routers.
5. Connect the routers with appropriate cables.
6. Assign IP addresses to the PCs and router interfaces.
7. Configure the router interfaces with their respective IP addresses.
8. Enable RIP on the routers.
9. Add the connected networks to RIP configuration.
10. Verify the routing table of each router.
11. Use the Ping command to test communication between different networks.

Observe that the routers automatically learn the routes through RIP. **RIP**

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Learning Outcome

After completing this experiment, the learner will be able to:

1. **Understand the concept of dynamic routing** and its role in enabling communication between multiple networks by automatically learning and updating routing information.
2. **Develop practical skills in configuring network devices** by assigning IP addresses, subnet masks, default gateways, and router interfaces in Cisco Packet Tracer.